Dual-Teacher Prototype Consistency for Unsupervised Domain Adaptive Object Detection in Remote Sensing Images

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***Abstract*—Unsupervised domain-adaptive object detection (UDA-OD) in remote sensing images aims to transfer knowledge from a labeled source domain to an unlabeled target domain, mitigating performance degradation caused by domain shift. Existing methods commonly adopt a teacher–student framework, where a single teacher, updated via an exponential moving average (EMA) of the student parameters, generates pseudolabels for training. While this framework effectively transfers knowledge from the teacher to the student, the quality of pseudolabels and the alignment of source- and target-domain features remain key challenges for reducing domain shift. To address these challenges, a dual-teacher prototype consistency (DTPC) method is proposed. Specifically, DTPC employs two available object detection models as teachers to provide complementary guidance for the student. The larger teacher, equipped with a deeper backbone and kept frozen, extracts domain-invariant prototype anchors through a class-aware prototype update (CAPU) mechanism, thereby facilitating effective feature-level alignment between the source and target domains. Furthermore, the larger teacher and an EMA-updated smaller teacher jointly generate pseudolabels to supervise the student with soft- and hard-label losses, improving the quality and reliability of pseudolabels. Extensive experiments on two remote sensing benchmarks demonstrate that DTPC achieves the state-of-the-art (SOTA) results, surpassing the second-best methods by 4.6% and 4.3% mean average precision (mAP) on the NWPU VHR-10 → detectIon in optical remote sensing images (DIOR) and VisDrone → DroneVehicle adaptation settings, respectively.**

***Index Terms*—Contrastive learning, object detection, remote sensing images, unsupervised domain adaptation (UDA).**

I. Introduction

DEEP learning-based object detection algorithms have been extensively applied in remote sensing images, covering various applications, such as traffic monitoring [1], natural disaster response [2], and building detection [3]. However, the detection performance of these models tends to degrade significantly when there exists a large distribution discrepancy between the training and testing datasets. To address this issue, unsupervised domain-adaptive object detection (UDA-OD) has been proposed [4], [5], [6], [7], [8], which transfers knowledge from a labeled source domain to an unlabeled target domain, aiming to alleviate performance degradation caused by domain shift.

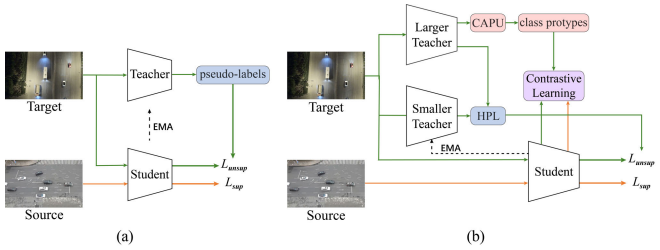


Fig. 1. Comparison with the teacher–student framework. (a) Teacher–student. (b) Proposed DTPC uses dual teachers to improve feature-level alignment and enhance pseudolabel reliability.

Mainstream UDA-OD methods are based on the teacher–student framework [9], as illustrated in Fig. 1(a). In this framework, the teacher network generates pseudolabels for the unlabeled target domain to supervise the student network. The student network is trained using labeled source data and pseudo-labeled target data, generating predictions for both domains. Teacher parameters are updated through an exponential moving average (EMA) [10] of the student, which maintains temporal consistency and stabilizes training. Existing methods design pseudo-label selection strategies to enhance pseudolabel quality [11], [12]. Other methods employ adversarial training to extract domain-invariant feature representations [4], [13], while some utilize prototype-based contrastive learning to mitigate feature discrepancies between source and target domains [14]. Hard-label loss is widely adopted in the aforementioned methods [7], [15]. However, this amplifies noise sensitivity and weakens domain adaptation performance when pseudolabels are inaccurate. Furthermore, while prototype-based contrastive learning aligns source and target feature spaces using class prototypes [16], it typically ignores domain-invariant features and features from minority classes, reducing adaptation performance.

To further improve pseudolabel quality and domain feature alignment, a dual-teacher prototype consistency (DTPC) framework is proposed for UDA-OD in remote sensing images. DTPC employs two distinct teacher networks with complementary architectures to guide the student, as illustrated in Fig. 1(b). The larger teacher, with a deeper backbone, can extract high-level semantic features that are less sensitive to domain shifts [17], [18]. These features are used to con-

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struct domain-invariant prototype anchors via a class-aware prototype update (CAPU) mechanism, guiding the student to align source and target features through contrastive learning. Simultaneously, a hybrid pseudolabel loss (HPL) mechanism integrates predictions from the EMA-updated smaller teacher and the larger teacher using a combination of soft- and hard-label losses, enhancing pseudolabel reliability and quality.

In this article, the proposed CAPU mechanism aims to extract class prototypes from the larger teacher network and utilize them as domain-invariant anchors for feature-level alignment. Particularly, CAPU considers the inherent class imbalance problem commonly observed in remote sensing datasets. It dynamically adjusts the confidence threshold and the prototype update momentum based on the occurrence frequency of each class. Specifically, features exceeding the thresholds are aggregated to form class prototypes. During prototype updates, classes with lower occurrence frequency, referred to as minority classes, are updated with smoother momentum to maintain representative prototypes under limited samples. The resulting prototypes serve as domain-invariant anchors, which guide the student model to align source and target features through contrastive learning, thereby alleviating domain discrepancy and enhancing interclass separability. The HPL mechanism enhances both the quality and reliability of pseudolabels by selectively integrating soft and hard labels. Hard labels are adopted only when predictions from the two teacher models are consistent, and the confidence of the larger teacher exceeds a predefined threshold. This can reduce noise from incorrect or low-confidence predictions and improve pseudolabel accuracy. In uncertain or low-confidence regions, soft labels from the smaller teacher are retained to prevent instability during training. By combining high-quality hard labels with robust soft labels, HPL increases the overall reliability of pseudolabels and supports more stable and accurate student model optimization.

The main contributions of this article are as follows.

1) A CAPU mechanism is designed to construct domain-invariant anchors by adaptively balancing class frequencies, thereby enabling explicit feature-level alignment across domains.

2) An HPL mechanism is developed to adaptively combine soft- and hard-label losses based on the predictions of the two teacher networks, effectively improving the quality of pseudolabels during training.

3) The proposed DTPC framework on two remote sensing benchmarks demonstrates superior performance and robustness.

Sections II–V of this article are structured as follows. Section II reviews the related work. Section III presents the details of the proposed method and its implementation. Section IV reports the experimental results and analysis. Finally, Section V concludes this work.

II. Relate Work

A. Unsupervised Domain-Adaptive Object Detection

Unsupervised domain adaptation (UDA) aims to mitigate the performance degradation caused by domain shift between the labeled source domain and the unlabeled target domain. In object detection, existing approaches can be broadly categorized into generative-based methods, adversarial-based methods, and self-training methods.

Generative-based methods align cross-domain feature distributions by generating domain-transferred images. The most prevalent strategy employs unpaired image-to-image translation algorithms, such as Cycle-GAN [19], unsupervised image-to-image translation (UNIT) [20], and multimodal UNIT [21]. Several recent works [13], [22] translate source (or target) images into the target (or source) domain while optimizing adversarial or consistency losses in the input or output spaces to reduce domain discrepancies.

Adversarial-based methods minimize domain discrepancies through domain discriminators to align feature distributions across domains. Chen et al. [4] pioneered this line of work by introducing dual-level feature alignment, inspiring many subsequent approaches [5], [6]. While demonstrating effectiveness, adversarial training suffers from inherent instability and often overlooks category-aware semantics. To address these issues, recent works [23], [24] have incorporated contrastive learning loss optimization, which is also adopted in our method for feature-level distribution alignment.

Self-training methods utilize generated pseudolabels for supervised training to adapt source-trained models to target domains. Early works [11], [25] primarily relied on confidence-based pseudolabel filtering strategies to suppress noise, while subsequent studies introduced teacher–student framework [26], consistency regularization [7], and uncertainty modeling [27] to further improve pseudolabel quality and training stability. Hybrid strategies that combine self-training with adversarial alignment [28], [29] have also been explored. Recent studies [15], [16], [30] place greater emphasis on pseudolabel noise suppression and multidimensional optimization. Unlike previous works, our approach leverages not only response-level pseudolabels but also larger teacher guidance in the feature space to enhance adaptation.

B. Remote Sensing UDA

Although UDA methods have achieved progress on natural images, they remain challenging in remote sensing images due to complex scenes, scale variations, and diverse backgrounds. One major difficulty lies in feature alignment across domains [31]. To address this, Xu et al. [5] introduced FADA, which employs adversarial foreground alignment and prototype confusion to mitigate domain shifts, while Han et al. [32] further enhanced stability under long-tailed distributions by incorporating frequency-based feature alignment and adaptive pseudolabel assignment. Another key issue is the presence of noisy pseudolabels and strong background interference. To alleviate these problems, Zhu et al. [6] developed RFA-Net, leveraging feature reconstruction with pseudolabel constraints at the instance level, and Liu et al. [33] proposed CR-Mixing, which integrates region-level mixing with adversarial learning under a mean-teacher framework to suppress label noise and strengthen foreground representations. Beyond alignment and noise suppression, insufficient intraclass consistency remains a bottleneck. Zhang et al. [34] tackled this with FIE-Net, which

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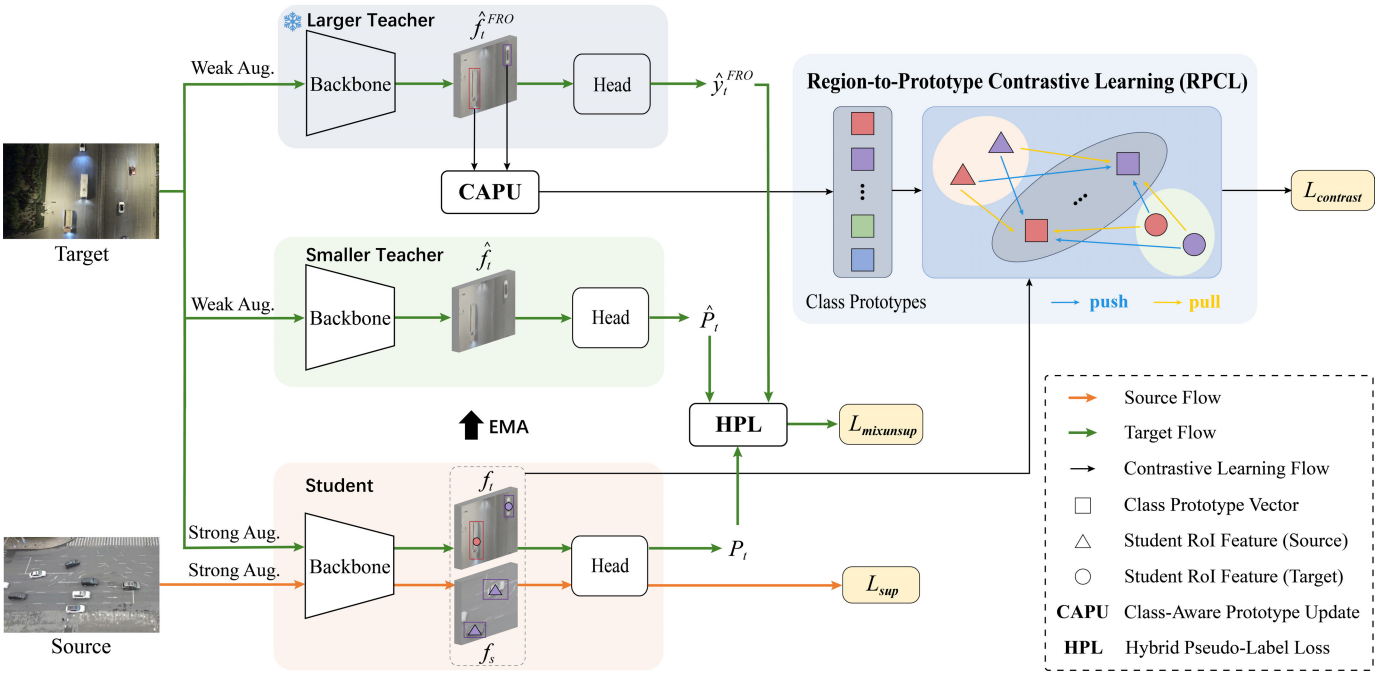


Fig. 2. Overall framework of the proposed DTPC method. The framework employs a dual-teacher paradigm, where the student detector is guided by two complementary teacher networks with distinct architectures. The frozen larger teacher provides semantically rich representations that capture domain-consistent characteristics for prototype construction, while the EMA-updated smaller teacher offers fine-grained and robust supervision. The CAPU mechanism adaptively adjusts confidence thresholds and momentum coefficients based on category frequency to continuously refine class prototypes. These prototypes, serving as domain-invariant anchors, are integrated into the RPCL to guide the student model in achieving feature-level alignment, and the HPL mechanism adaptively fuses soft- and hard-label losses based on class agreement and confidence to enhance the stability and accuracy of pseudolabel supervision.

improves both intraclass consistency and foreground discriminability through multigranularity instance augmentation and mixed-threshold pseudolabel selection.

These studies demonstrate that cross-domain object detection has made significant progress in remote sensing images. However, existing methods remain vulnerable to noisy pseudolabels, class imbalance, and unstable feature alignment, which limit their overall performance. Therefore, the DTPC is proposed to address these issues and improve feature alignment robustness, thus further improving model performance.

C. Contrastive Learning for Domain Adaptation

Contrastive learning aims to learn discriminative feature representations by constructing positive and negative pairs, pulling together similar samples while pushing apart dissimilar ones. In UDA for object detection, it has proven effective for feature modeling and mitigating domain shift, while existing approaches can be roughly categorized into instance-based and region-based methods.

Instance-based methods [35], [36], [37] primarily treat individual images or object instances as contrastive units. They construct positive pairs using different augmented views of the same instance while treating other samples as negatives, performing pixel- or instance-level contrast to learn discriminative global features. Representative designs include momentum encoders with dynamic queues [35], [36], and memory-bank-based cross-batch contrastive learning [37].

Region-based methods [38], [39] focus on semantic regions, such as object proposals, pseudolabeled regions, or class prototypes. By contrasting regions of the same category, they enhance semantic consistency and discriminability. For instance, domain contrast (DC) [40] mitigated category bias via cross-domain instance-level contrast, category contrast (CaCo) [41] introduced a category-aware dictionary to achieve semantic center alignment, and CMT [30] leveraged teacher-generated pseudolabeled regions to construct contrastive pairs, improving robustness against noisy pseudolabels. Our method also adopts a similar region-level enhancement strategy.

Building upon region-level contrast, memory modules have been introduced to store cross-domain category prototypes or semantic features, enhancing feature modeling and domain alignment. MILA [14] stored source-domain instance features in memory and retrieved the most similar source instances for target samples to achieve cross-domain alignment. DATR [16] accumulated dataset-level prototypes across domains in memory and applies contrastive loss to improve global representation and interclass separability. Distinct from previous works, our method leverages a larger teacher model to generate pseudolabels for filtering high-quality target-domain instance features, which are then updated via a class-aware momentum mechanism to construct domain-invariant class prototypes for feature-level alignment.

III. Proposed Method

The DTPC framework is shown in Fig. 2 as an overview. Under a dual-teacher paradigm, the student detector is jointly guided by two complementary teacher networks with distinct architectures. The larger teacher, equipped with a deeper backbone and pretrained on the source domain, remains frozen

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during adaptation and provides semantically rich and domain-invariant representations across diverse scenes. Meanwhile, the EMA-updated smaller teacher offers fine-grained and dynamic supervision. To fully exploit the features extracted by the larger teacher, the CAPU mechanism adaptively adjusts confidence thresholds and momentum coefficients to enhance the representation of minority classes and construct reliable class prototypes as domain-invariant anchors. These prototypes are then employed by the region-to-prototype contrastive learning (RPCL), which guides the student model to align source and target features with the domain-invariant anchors for feature-level alignment, while enhancing interclass feature separability. Furthermore, the HPL mechanism adaptively fuses soft- and hard-label supervision based on class agreement and confidence, thereby enhancing pseudolabel reliability and model discriminability.

A. Class-Aware Prototype Update

In some prototype-based contrastive learning methods [42], [43], [44], prototypes are dynamically updated during training through confidence-based filtering to exclude low-quality features, followed by momentum updates to maintain stability. The prototype update is expressed as

$$\mu_c \leftarrow m \cdot \mu_c + (1-m) \cdot \bar{f}_c \tag{1}$$

where μ_c is the prototype of class c, $\bar{f}_c$ is the mean of region-of-interest (ROI) features that are predicted as class c with confidence scores above the threshold in the current iteration, and m is the globally shared momentum coefficient.

This strategy of using a globally fixed confidence threshold and momentum coefficient fails to account for the inherent class imbalance in remote sensing datasets, where minority classes contain fewer samples and produce low-confidence pseudolabels. Consequently, many informative features from rare classes are discarded due to the use of a high fixed confidence threshold, while a shared momentum coefficient limits the adaptability of prototypes across categories. These limitations lead to unstable and biased prototype representations, weakening class discrimination and domain alignment.

To mitigate these problems, the CAPU mechanism is proposed, which dynamically adjusts classwise confidence thresholds and momentum coefficients based on category frequency. As illustrated in Fig. 2, it enables the construction of higher-quality prototypes for RPCL, and the procedure is summarized in Algorithm 1. Concretely, for minority classes in early training stages, CAPU adopts lower confidence thresholds to admit more features into prototype construction and applies smaller momentum coefficients to enhance the responsiveness of the prototype to new features. In contrast, classes with higher occurrence frequency are treated more conservatively to prevent them from dominating the feature space.

The class frequency ratio p_c is first defined as

$$p_c = \frac{n_c}{\sum_{i=1}^{C} n_i} \tag{2}$$

where n_c denotes the running count of samples assigned to class c based on pseudolabel statistics collected during training, and C is the total number of classes. This frequency ratio is used to measure the degree of class imbalance; a lower value of p_c indicates that the class is less frequent.

Algorithm 1 Class-Aware Prototype Update

Input: Unlabeled target batch X_t, initial prototypes $\boldsymbol{\mu}^0 = \{\mu_c^0\}_{c=1}^C$, base momentum m_{base}, base confidence threshold θ_{base}, scaling factors β_m, β_θ, lower bounds $m_{\min}, \theta_{\min}$
Output: Continuously updated prototypes $\boldsymbol{\mu} = \{\mu_c\}_{c=1}^C$

for each class $c = 1$ **to** C **do**
  $n_c \leftarrow 0$
  $\mu_c \leftarrow \mu_c^0$
end for
for each iteration **do**
  Obtain ROI feature set $\mathcal{F} = \{f_i\}_{i=1}^{N_r}$ with predicted class labels $\{l_i\}_{i=1}^{N_r}$ and confidence scores $\{s_i\}_{i=1}^{N_r}$ by forwarding X_t through the frozen larger teacher, where i indexes ROI instances and N_r denotes their total number in the current iteration.
  for each class $c = 1$ **to** C **do**
    Update class sample counts $n_c \leftarrow n_c + |\{i \mid l_i = c\}|$
  end for
  for each class $c = 1$ **to** C **do**
    Compute class frequency ratio p_c by (2)
    Compute classwise confidence threshold θ_c by (3)
    Compute classwise momentum coefficient m_c by (4)
    Select ROI feature set $\mathcal{R}_c = \{f_i \mid l_i = c,\ s_i \geq \theta_c\}$
    if $\mathcal{R}_c \neq \emptyset$ **then**
      Compute mean feature $\bar{f}_c = \frac{1}{|\mathcal{R}_c|} \sum_{f_i \in \mathcal{R}_c} f_i$
      Update $\mu_c \leftarrow m_c \cdot \mu_c + (1 - m_c) \cdot \bar{f}_c$
    end if
  end for
end for

Building upon this, the class-adaptive confidence threshold θ_c and momentum coefficient m_c are defined as explicit functions of the class frequency

$$\theta_c = \max\left(\theta_{\text{base}} - \beta_\theta \cdot (1 - p_c),\ \theta_{\min}\right) \tag{3}$$

$$m_c = \max\left(m_{\text{base}} - \beta_m \cdot (1 - p_c),\ m_{\min}\right) \tag{4}$$

where θ_{base} and m_{base} denote the base confidence threshold and base momentum coefficient, respectively. The scaling factors β_θ and β_m control the sensitivity to class frequency, while the lower bounds $\theta_{\min}$ and $m_{\min}$ prevent overly noisy feature inclusion and ensure stable prototype updates. In our implementation, we set $\beta_\theta = 0.1$ and $\theta_{\min} = 0.8$ for the confidence threshold and $\beta_m = 0.01$ and $m_{\min} = 0.95$ for momentum.

Although θ_c and m_c share a similar class-aware formulation, they play complementary roles in the prototype update process. For classes with lower frequency, the term $(1 - p_c)$ becomes larger, resulting in a greater reduction from the corresponding base values. Consequently, they are assigned lower confidence thresholds and smaller momentum coefficients.

To ensure the effectiveness of contrastive learning, prototypes are initialized by averaging classwise ROI features

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extracted by the larger teacher from 100 randomly sampled target images to form representative semantic centers, formulated as

$$\mu_c^0 = \frac{1}{\left|\mathcal{F}_c^0\right|} \sum_{f_i \in \mathcal{F}_c^0} f_i \tag{5}$$

where $\mathcal{F}_c^0$ denotes the set of ROI features predicted as class c by the frozen larger teacher during initialization, and f_i represents an ROI feature in this set.

B. Region-to-Prototype Contrastive Learning

With the refined prototypes generated by the CAPU as domain-invariant anchors, RPCL is further employed to guide the student model in aligning source and target features. Unlike previous approaches that construct class prototypes from dynamically updated EMA teacher features, RPCL builds more reliable class prototypes from the domain-consistent representations of the frozen larger teacher and aligns region-level features of both source and target domains with their corresponding prototypes via contrastive learning. This design facilitates more robust cross-domain representation learning while enhancing intraclass compactness and intra-class separability.

Specifically, as shown in Fig. 2, ROI features are first extracted from target domain images by the frozen larger teacher, along with their predicted class labels and confidence scores, and aggregated by class to form class prototypes. Concurrently, the student model extracts ROI features from both source and target domains and aligns them with the corresponding class prototypes via both intradomain and cross-domain contrastive learning. This strategy improves the consistency of feature representations through class-level aggregation and establishes cross-model and cross-domain alignment, thereby enhancing feature discriminability and generalization capability.

The standard InfoNCE [45] loss is employed for contrastive optimization. The intradomain contrastive loss is computed within the target domain, encouraging ROI features f_i^t to align with their corresponding prototype $\mu_{l_i}^t$, and is defined as

$$\mathcal{L}_{\text{intra}} = -\log \frac{\exp\left(\text{sim}\left(f_i^t, \mu_{l_i}^t\right)/\tau\right)}{\sum_c \exp\left(\text{sim}\left(f_i^t, \mu_c^t\right)/\tau\right)} \tag{6}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity with ℓ_2 normalization, l_i is the class label of the ith ROI, and τ is the temperature parameter.

Similarly, the cross-domain contrastive loss is defined as $\mathcal{L}_{\text{cross}}$, which is computed between source-domain ROI features and target-domain class prototypes to promote feature alignment across domains. Finally, the overall contrastive objective combines intradomain and cross-domain losses

$$\mathcal{L}_{\text{contrast}} = \frac{1}{2}\left(\mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{cross}}\right). \tag{7}$$

C. Hybrid Pseudolabel Loss

The HPL mechanism is designed to enhance both the quality and reliability of pseudolabels by selectively integrating the outputs from the two teacher networks. Soft labels from the EMA-updated smaller teacher retain uncertainty, improving robustness and stabilizing early student training [46], [47], [48]. However, when used independently, soft labels are prone to oversmoothing, leading to ambiguous supervision. The soft label loss transfers the probabilistic knowledge from the smaller teacher to the student, and is computed from both the RPN and ROI heads of the two-stage detector separately

$$\mathcal{L}_{\text{soft}}^{\text{rpn}}\left(a_j\right) = \mathcal{L}_{\text{cls}}^{\text{rpn}}\left(p_j, \hat{p}_j^{\,smaller}\right) + \mathcal{L}_{\text{reg}}^{\text{rpn}}\left(t_j, \hat{t}_j^{\,smaller}\right) \tag{8}$$

$$\mathcal{L}_{\text{soft}}^{\text{roi}}\left(r_i\right) = \mathcal{L}_{\text{cls}}^{\text{roi}}\left(p_i, \hat{p}_i^{\,smaller}\right) + \mathcal{L}_{\text{reg}}^{\text{roi}}\left(t_i, \hat{t}_i^{\,smaller}\right) \tag{9}$$

where a_j is the j-th anchor in the RPN head and r_i is the i-th ROI instance in the ROI head. Here, p represents the predicted objectness scores associated with anchors or the predicted category probabilities associated with ROI instances, while t denotes the corresponding regression offsets. Similarly, $\hat{p}^{\,smaller}$ and $\hat{t}^{\,smaller}$ are the corresponding predictions generated by the smaller teacher.

In contrast, hard labels generated by the frozen larger teacher provide high-confidence category predictions and precise bounding boxes, offering stronger supervision and clearer decision boundaries. Nonetheless, solely using hard labels risks propagating errors from low-confidence or incorrect predictions. The hard-label loss is only applied to the ROI head in our method, where the student is supervised by the high-confidence predictions from the larger teacher

$$\mathcal{L}_{\text{hard}}^{\text{roi}}\left(r_i\right) = \mathcal{L}_{\text{cls}}^{\text{roi}}\left(p_i, \hat{y}_i^{\,larger}\right) + \mathcal{L}_{\text{reg}}^{\text{roi}}\left(t_i, \hat{t}_i^{\,larger}\right) \tag{10}$$

where $\hat{y}_i^{\,larger}$ and $\hat{t}_i^{\,larger}$ denote the one-hot class label and regression offsets predicted by the larger teacher, respectively.

Methods that rely solely on either soft- or hard-label losses fail to achieve an optimal balance between robustness and discriminative supervision. To overcome these issues, the HPL mechanism adaptively fuses the two types of losses based on class agreement and the confidence of the larger teacher, improving pseudolabel quality and reliability. The overall procedure is summarized in Algorithm 2.

During each training iteration, region proposals are generated by the RPN of the smaller teacher, transformed into ROI instances via RoIAlign, and shared by the student and the larger teacher for prediction. Since the smaller teacher is an EMA of the student, their RPN parameters remain highly consistent, ensuring spatial alignment while providing more stable proposals for reliable pseudo supervision. For each ROI instance r_i, when the predicted class labels $\hat{l}_i^{\,smaller}$ and $\hat{l}_i^{\,larger}$ are consistent, and the confidence score of the larger teacher $\hat{s}_i^{\,larger}$ exceeds the threshold θ_{conf} set to 0.8, the mixed pseudolabel loss in the ROI head is defined as

$$\mathcal{L}_{\text{mixunsup}}^{\text{roi}}\left(r_i\right) = \left(1 - \hat{s}_i^{\,larger}\right) \mathcal{L}_{\text{soft}}^{\text{roi}}\left(r_i\right) + \hat{s}_i^{\,larger}\, \mathcal{L}_{\text{hard}}^{\text{roi}}\left(r_i\right) \tag{11}$$

where the confidence score $\hat{s}_i^{\,larger}$ controls the fusion weight between soft and hard supervision, allowing the student model to dynamically adapt to pseudolabel reliability and balance robustness with discriminative ability.

If the predictions from the two teachers are inconsistent or the confidence of the larger teacher is below θ_{conf}, the hard labels are considered unreliable. In this case, only the soft

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Algorithm 2 Mixed Pseudolabel Loss Computation

Input: Unlabeled target batch X_t, confidence threshold θ_{conf}
Output: Mixed pseudolabel loss $\mathcal{L}_{\text{mixunsup}}$
Initialize $\mathcal{L}_{\text{rpn}} \leftarrow 0$, $\mathcal{L}_{\text{roi}} \leftarrow 0$
Obtain predefined anchors $\{a_j\}_{j=1}^{N_a}$ on the feature maps of X_t
Generate student RPN predictions for $\{a_j\}$: $\{p_j, t_j\}$
Generate smaller teacher RPN predictions for $\{a_j\}$: $\{\hat{p}_j^{smaller}, \hat{t}_j^{smaller}\}$
Obtain ROI instances $\{r_i\}_{i=1}^{N_r}$ via RoIAlign using the RPN proposals of the smaller teacher
Generate student ROI predictions for $\{r_i\}$: $\{p_i, t_i\}$
Generate smaller teacher ROI predictions for $\{r_i\}$: $\{\hat{p}_i^{smaller}, \hat{t}_i^{smaller}, \hat{l}_i^{smaller}\}$
Generate larger teacher ROI predictions for $\{r_i\}$: $\{\hat{y}_i^{larger}, \hat{t}_i^{larger}, \hat{l}_i^{larger}, \hat{s}_i^{larger}\}$
for $j = 1$ **to** N_a **do**
    Compute $\mathcal{L}_{\text{soft}}^{\text{rpn}}(a_j)$ by (8)
    $\mathcal{L}_{\text{rpn}} \leftarrow \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{soft}}^{\text{rpn}}(a_j)$
end for
for $i = 1$ **to** N_r **do**
    Compute $\mathcal{L}_{\text{soft}}^{\text{roi}}(r_i)$ by (9)
    if $\hat{l}_i^{smaller} = \hat{l}_i^{larger}$ **and** $\hat{s}_i^{larger} > \theta_{conf}$ **then**
        Compute $\mathcal{L}_{\text{hard}}^{\text{roi}}(r_i)$ by (10)
        $\mathcal{L}_{\text{roi}} \leftarrow \mathcal{L}_{\text{roi}} + (1 - \hat{s}_i^{larger})\mathcal{L}_{\text{soft}}^{\text{roi}}(r_i) + \hat{s}_i^{larger}\mathcal{L}_{\text{hard}}^{\text{roi}}(r_i)$
    else
        $\mathcal{L}_{\text{roi}} \leftarrow \mathcal{L}_{\text{roi}} + \mathcal{L}_{\text{soft}}^{\text{roi}}(r_i)$
    end if
end for
$\mathcal{L}_{\text{mixunsup}} \leftarrow \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{roi}}$

label loss is used to prevent potential noise from misleading the student

$$\mathcal{L}_{\text{mixunsup}}^{\text{roi}}(r_i) = \mathcal{L}_{\text{soft}}^{\text{roi}}(r_i). \tag{12}$$

The HPL mechanism is only applied to the ROI losses, while the RPN losses are always computed using soft distillation without mixing. The overall unsupervised loss is, therefore, computed as

$$\mathcal{L}_{\text{mixunsup}} = \sum_{j=1}^{N_a} \mathcal{L}_{\text{soft}}^{\text{rpn}}(a_j) + \sum_{i=1}^{N_r} \mathcal{L}_{\text{mixunsup}}^{\text{roi}}(\text{r}_\text{i}) \tag{13}$$

where N_a and N_r are the number of anchors and ROI instances in the current iteration, respectively.

D. Overall Objective

The total loss consists of a detection loss and a prototype contrastive loss. The detection loss includes both supervised and unsupervised components. The supervised loss is consistent with the standard detection loss used in the faster R-CNN [53] framework, while the unsupervised loss is defined as described in (11) and (12). Thus, the detection loss can be expressed as

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{sup}} + \lambda\mathcal{L}_{\text{mixunsup}} \tag{14}$$

where $\mathcal{L}_{\text{sup}}$ denotes the supervised detection loss on the source domain, and λ is a hyperparameter used to balance the supervised and unsupervised loss terms. The total loss is formulated as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + \eta\mathcal{L}_{\text{contrast}} \tag{15}$$

where $\mathcal{L}_{\text{contrast}}$ refers to the prototype contrastive loss in (7), and η controls the relative weight between the detection loss and the contrastive loss.

IV. Experiment

A. Datasets

1) NWPU VHR-10 [49]: This dataset comprises 800 images, including 715 high-spatial-resolution RGB images sourced from Google Earth and 85 pansharpened color–infrared images (CIR) from the Vaihingen dataset. The spatial resolution of the images from Google Earth ranges from 0.5 to 2 m, while that from the Vaihingen dataset is 0.08 m. Among the 800 high-resolution images, 650 are positive images with the available target in the annotation. In contrast, the rest of the images do not contain the target object, considered a negative image. In our experiment, only positive images from the dataset are used to form the source-domain dataset, which contains ten object categories. Table I compares all datasets used in our experiments.

2) DetectIon in optical remote sensing images (DIOR) [50]: This dataset comprises 23 463 RGB images of optical remote sensing scenes, containing a total of 192 472 objects categorized into 20 common classes. All images originate from Google Earth and exhibit spatial resolutions ranging between 0.5 and 3 m. Unlike the NWPU VHR-10 dataset, DIOR offers a substantially broader spectrum of image variations due to its global coverage spanning over 80 countries. This extensive diversity introduces significant variability within each class, reflected in changes of viewing angles, occlusions, visual appearances, backgrounds, object poses, and other factors. Ten categories corresponding to those in the NWPU VHR-10 dataset are extracted from DIOR to form DIOR*, which serves as the target-domain dataset. The DIOR* dataset comprises a total of 5862 images and 28 527 objects. In addition, we analyze the class frequency distribution of DIOR*, as illustrated in Fig. 3. The distribution exhibits evident class imbalance, with certain categories dominating the dataset. Such an imbalance can bias prototype learning in contrastive representation learning, leading to uneven feature optimization across categories and motivating mechanisms that regulate classwise prototype updates during adaptation.

3) VisDrone [51]: The VisDrone dataset consists of 10 209 images, including 6471 training images, 3190 testing images, and 548 validation images, covering ten object categories. These images were captured by various types of UAVs across multiple urban scenes under diverse weather and lighting conditions, exhibiting high variability and complexity. Four common object categories shared between the VisDrone dataset and the DroneVehicle dataset are extracted to construct VisDrone*, which serves as the source-domain dataset.

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TABLE I
DATASETS COMPARISON OF THE SOURCE AND TARGET DOMAINS

domain	dataset	data source	#classes	GSD (m)	image size	#images
D_s	NWPU VHR-10 [49]	GoogleEarth	10	0.08 ~ 2	601×381 ~ 1728×1028	800
D_t	DIOR [50]	GoogleEarth	20	0.5 ~ 3	800×800	23463
D_s	VisDrone [51]	UAV	10	–	480×360 ~ 2k×1.5k	10209
D_t	DroneVehicle [52]	UAV	5	–	640×512	56878

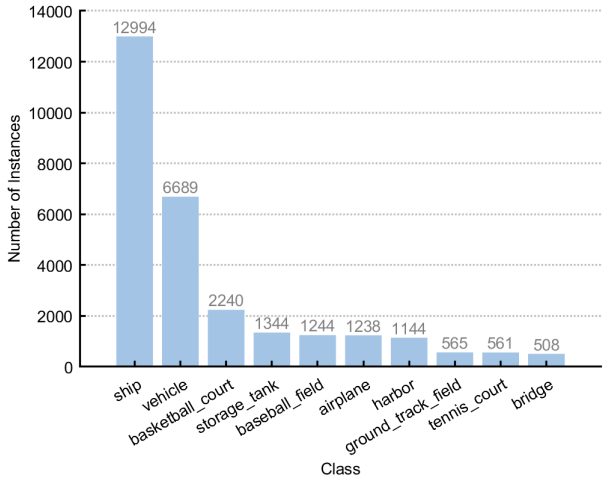


Fig. 3. Class frequency distribution of the DIOR* target dataset. The distribution shows a pronounced class imbalance, with substantial imbalance across categories.

4) DroneVehicle [52]: The DroneVehicle dataset is a large-scale UAV-based vehicle dataset consisting of 28 439 pairs of RGB and infrared images, captured under diverse conditions ranging from daytime to nighttime. Compared with the VisDrone dataset, DroneVehicle exhibits a higher degree of diversity in terms of scene distribution, object density, and environmental complexity, including various representative scenarios, such as urban roads, residential areas, parking lots, and under overpasses, as well as varying illumination conditions across different time periods. In our experiments, only the RGB images from the dataset are utilized. To ensure category consistency with the VisDrone dataset, four common object categories are extracted to construct a subset referred to as DroneVehicle*, which serves as the target-domain dataset.

B. Evaluation Metric

To quantitatively evaluate the detection performance of the proposed DTPC framework, standard evaluation metrics widely used in object detection tasks are employed: average precision (AP) and mean average precision (mAP). The AP is defined as the area under the precision–recall curve, integrating both precision and recall, and is expressed as

$$\mathrm{AP} = \int_0^1 P(R)\, dR \tag{16}$$

where P represents the precision rate and R is the recall rate. The mAP is computed by averaging AP values across all categories

$$\mathrm{mAP} = \frac{1}{c}\sum_{i=1}^{c} \mathrm{AP}_i \tag{17}$$

where c is the total number of categories.

C. Implementation Details

The faster R-CNN [53] architecture with VGG16 [57] as the backbone is adopted for feature extraction. Experiments are conducted using source and target images with a batch size of 16, utilizing two NVIDIA GeForce RTX 4090 GPUs. The learning rate is set to 0.02 without any decay schedule. During the burn-in stage, the model is initialized with labeled source samples for 10K iterations, and an EMA-updated copy of the model parameters is saved and used to initialize both the student and teacher networks. The training then continues jointly on the source and target domains for an additional 30K iterations. Furthermore, the larger teacher is pretrained on labeled source samples using a ConvNeXt-FPN [58], [59] backbone for 10K iterations.

For the teacher–student framework, the experimental setup proposed by Li et al. [7] is followed. Our weak data augmentation includes random horizontal or vertical flips, while strong augmentation consists of color jittering, gray scale, Gaussian blur, and cutout [60]. The teacher network is updated from the student network using EMA at each iteration, with the EMA momentum α set to 0.9996 for progressive updates. In addition, θ_{base} in (3) and m_{base} in (4) are set to 0.95 and 0.99 as the initial confidence threshold and initial prototype update momentum, respectively. The parameters λ in (14) and η in (15) are set to 2 and 3, respectively.

D. Experimental Results and Analyses

In this section, the DTPC method is evaluated both qualitatively and quantitatively for object detection in remote sensing images. As described in Section IV-A and summarized in Table I, four remote sensing datasets are selected to conduct our experiments. DTPC is compared with recent domain adaptation object detectors under two adaptation settings. Since most existing methods focus on natural scenes, they are reimplemented for the remote sensing scenario using their official implementations and default configurations. In addition, two baseline models are introduced for comparison. The source-only model is trained solely on labeled source-domain data and evaluated on the target domain, while the oracle model is trained and tested using labeled target-domain data, serving as an upper performance bound. The detailed experimental settings and results are presented as follows.

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TABLE II

QUANTITATIVE COMPARISON OF THE NWPU VHR-10 TO DIOR* EXPERIMENTS WITH THE SOURCE-ONLY, ORACLE, AND OTHER DOMAIN ADAPTATION OBJECT DETECTION APPROACHES. ALL METHODS USE VGG16 AS THE BACKBONE

NWPU VHR-10 → DIOR*												
Methods	Detector	Airplane	Ship	Storage tank	Baseball field	Tennis court	Basketball court	Ground track field	Harbor	Bridge	Vehicle	mAP
Source only	FRCNN	35.9	3.6	1.6	62.0	62.8	42.3	28.9	1.6	4.6	29.0	27.2
Oracle		80.1	77.3	76.9	89.8	95.2	93.0	75.6	53.7	43.6	71.5	75.7
SIGMA [54]	FCOS	46.2	6.2	13.3	65.2	82.9	67.2	71.9	18.6	9.9	32.1	41.4
HT [55]	FCOS	64.1	**22.6**	16.9	52.8	89.3	80.5	71.6	16.4	**20.9**	34.4	47.0
FIE-Net [34]	FCOS	64.3	11.7	15.4	85.1	90.0	80.6	71.5	14.7	18.1	37.2	48.9
AT [7]	FRCNN	51.8	7.6	15.1	69.5	89.9	70.5	63.2	15.7	8.0	37.5	42.9
PT [27]	FRCNN	64.7	7.2	12.5	65.0	92.2	77.5	59.0	15.5	12.1	39.4	44.5
CMT [30]	FRCNN	68.6	8.1	20.2	64.6	94.1	81.8	64.4	17.9	14.6	38.2	47.3
2PCNet [8]	FRCNN	41.6	6.5	10.9	59.3	86.4	72.9	65.1	13.9	5.1	29.0	39.1
ALDI++ [56]	FRCNN	71.2	13.6	19.9	84.7	95.8	86.7	70.4	**20.2**	8.5	40.6	51.2
Ours(DTPC)	FRCNN	**71.8**	17.9	**42.3**	**88.4**	**97.6**	**94.5**	**72.3**	15.6	14.2	**43.7**	**55.8**

1) NWPU VHR-10 to Dior:* The domain adaptation task from NWPU VHR-10 to Dior* is first investigated. The experimental results are summarized in Table II. Compared with the source-only baseline, our proposed DTPC method improves AP by 28.6%, demonstrating its effectiveness in addressing domain shift in cross-domain object detection. As shown in Table II, DTPC also outperforms all listed domain adaptation methods and achieves the current best performance. Existing approaches based on the mean teacher framework, such as AT [7], PT [27], and 2PCNet [8], achieve 42.9%, 44.5%, and 39.1% mAP, respectively, but their performance remains limited by the instability introduced by noisy pseudolabels. Building upon this paradigm, CMT [30] leverages contrastive learning to mitigate pseudolabel noise, reaching 47.3% mAP. One-stage FCOS-based approaches, including SIGMA [54], HT [55], and FIE-Net [34], report 41.4%, 47.0%, and 48.9% mAP, respectively, with improvements in certain categories but limited overall gains. The latest ALDI++ [56] achieves a further improvement to 51.2% mAP through a two-stage distillation mechanism. Our approach achieves an overall mAP of 55.8%, surpassing the second-best result by 4.6% on this benchmark. Notably, DTPC shows particularly significant improvements on categories that are more susceptible to domain discrepancies, such as storage tanks and baseball fields, effectively alleviating category bias and enhancing feature alignment across domains.

2) VisDrone to DroneVehicle*:* Table III reports the experimental results for the VisDrone* → DroneVehicle* domain adaptation task. Among the listed domain adaptation methods, DTPC achieves the best performance with an mAP of 44.1%, which is 25.4% points higher than the source-only baseline. Due to the presence of low-light and nighttime scenes in the DroneVehicle* dataset, methods that rely on single-feature representations and are sensitive to illumination variations, such as AT [7] and CMT [30], demonstrate relatively poor performance. In contrast, approaches emphasizing diversified feature fusion and uncertainty modeling, like HT [55] and PT [27], as well as methods specifically designed for day-to-night scenarios, such as 2PCNet [8], exhibit greater robustness and achieve better results under these challenging lighting conditions.

TABLE III

QUANTITATIVE COMPARISON OF THE VISDRONE* TO DRONEVEHICLE* EXPERIMENTS WITH THE SOURCE-ONLY, ORACLE, AND OTHER DOMAIN ADAPTATION OBJECT DETECTION APPROACHES. ALL METHODS USE VGG16 AS THE BACKBONE

VisDrone* → DroneVehicle*						
Methods	Detector	Car	Truck	Bus	Van	mAP
Source only	FRCNN	48.6	5.1	17.0	4.1	18.7
Oracle		79.0	44.6	76.7	27.4	56.9
SIGMA [54]	FCOS	62.1	15.2	41.8	5.2	31.1
HT [55]	FCOS	67.9	23.8	46.8	4.5	35.8
FIE-Net [34]	FCOS	64.7	8.6	42.6	9.7	31.4
AT [7]	FRCNN	44.1	9.3	40.1	3.5	24.3
PT [27]	FRCNN	**70.8**	14.0	31.5	5.2	30.4
CMT [[30]	FRCNN	49.6	14.0	31.9	3.4	24.7
2PCNet [8]	FRCNN	60.2	18.7	37.1	5.8	30.5
ALDI++ [56]	FRCNN	66.7	31.3	51.1	10.1	39.8
Ours(DTPC)	FRCNN	69.9	**37.1**	**57.3**	**12.2**	**44.1**

E. Ablation Study

In this section, an ablation study is first conducted to evaluate the effectiveness of each component by demonstrating its contribution to the overall performance. Furthermore, differences between prototype contrastive learning guided by the EMA-updated smaller teacher and the frozen larger teacher are analyzed, along with the impact of using different backbones for the larger teacher. Finally, several key hyperparameters are analyzed, including the initial base confidence θ_{base} in (3), the initial base momentum m_{base} in (4), the unsupervised weight λ

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TABLE IV
ABLATION STUDY OF THE COMPONENTS IN OUR PROPOSED DTPC

Method	RPCL	CAPU	HPL	mAP
baseline				51.2
Proposed	✓			53.5
	✓	✓		54.6
			✓	52.4
	✓		✓	54.8
Ours	✓	✓	✓	**55.8**

in the detection loss $\mathcal{L}_{det}$ in (14), and the contrastive weight η in the total loss $\mathcal{L}_{total}$ in (15). The benchmark of NWPU VHR-10 to DIOR* is utilized for conducting the ablation study.

1) Effectiveness of Each Component: To verify the effectiveness of each component in DTPC, individual modules of RPCL, CAPU, and HPL are equipped separately. Starting from the baseline, each module is progressively introduced, and the corresponding performance changes are observed. To specifically demonstrate the effect of CAPU, RPCL is applied alone using fixed initial confidence thresholds and momentum coefficients without dynamic adjustment. The baseline model adopts the soft-label mean teacher framework. The experimental results are summarized in Table IV.

From the results, it is observed that equipping with any single module leads to performance improvement, but the gains are relatively limited. Specifically, when RPCL is introduced alone, the mAP increased from 51.2% of the baseline to 53.5%, achieving an improvement of 2.3%. Since RPCL is the first component that introduces the additional teacher into the framework, this gain reflects the combined effect of teacher supervision and prototype-based contrastive learning. Although the larger teacher provides semantically rich and domain-consistent representations for prototype construction, the performance improvement remains moderate and is still constrained by the instability of prototype updates under class imbalance. Similarly, when HPL is employed alone, the mAP reaches 52.4%, only 1.2% higher than the baseline. This indicates that although the hybrid strategy of soft and hard labels improved the quality and reliability of pseudolabels to some extent, the model still suffers from insufficient feature alignment in the absence of high-quality prototype guidance.

When RPCL and HPL are combined, the model achieves a significant performance gain of 3.4% over the baseline. The CAPU module facilitates the construction of more representative prototypes for minority classes, thereby enhancing the effectiveness of cross-domain feature alignment. Finally, when all three modules are integrated, the model achieves an mAP of 55.8% on the target domain, representing a 4.6% improvement over the baseline. Compared with the 2.3% gain brought by RPCL alone, this result suggests that a substantial portion of the overall improvement arises from the proposed CAPU and HPL mechanisms rather than solely from the introduction of the additional teacher, demonstrating the synergistic effect of the proposed components.

TABLE V
RESULTS OF MODEL PERFORMANCE UNDER DIFFERENT TEACHER FEATURE SOURCES

Feature Source	Backbone	mAP	Δ
-	-	51.2	-
Smaller teacher	VGG16	53.0	+1.8
Frozen teacher	VGG16	53.5	+2.3
Larger teacher	ConvNeXt-FPN	**55.8**	+4.6

TABLE VI
RESULTS OF MODEL PERFORMANCE UNDER DIFFERENT TEACHER MODEL SIZES

Larger Teacher Backbone	mAP
ResNet50	54.0
ResNet101	54.4
ResNet152	54.5
ResNet50-FPN	54.7
ConvNeXt-FPN	**55.8**

2) Effectiveness of Different Teacher Feature Sources: In this part, the impact of different teacher feature sources used in prototype contrastive learning is analyzed. The experimental results are summarized in Table V. Specifically, the smaller teacher shares the same backbone as the student model and is updated via EMA during training, while the frozen teacher uses the same backbone but remains fixed. In addition, the larger teacher adopts a deeper backbone (ConvNeXt-FPN) and is also kept frozen during training.

Results demonstrate that the frozen teacher yields better performance than the smaller teacher, improving mAP by 0.5%. This confirms that the frozen teacher, by extracting features with domain-invariant characteristics and semantic consistency, provides a more stable alignment reference in the feature space, thereby enhancing cross-domain feature consistency. Furthermore, leveraging a larger teacher with a more powerful backbone brings an additional performance gain, achieving a 4.6% improvement over the baseline. This highlights the benefit of transferring richer and more representative features from high-capacity teacher models.

3) Effectiveness of Different Teacher Model Sizes: Following the previous experiment that demonstrated the effectiveness of a larger teacher in guiding prototype contrastive learning, the influence of the capacity of the larger teacher model on cross-domain adaptation performance is further examined. The results are summarized in Table VI.

Under the same student model, various larger teacher architectures are compared. When gradually increasing the teacher depth from ResNet-50 to ResNet-152, the model achieves an mAP improvement from 54.0% to 54.5%. This indicates that deeper networks can provide more representative and expressive features, leading to more effective prototype construction. However, the performance gain between ResNet-101 and ResNet-152 is relatively small, suggesting diminishing returns in contrastive learning performance as teacher depth increases beyond a certain point.

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TABLE VII

PERFORMANCE OF PROTOTYPE UPDATES WITH DIFFERENT INITIAL CONFIDENCE THRESHOLDS θ_{base} AND INITIAL MOMENTUM m_{base}

θ_{base} \ m_{base}	0.97	0.98	0.99	0.999
0.85	52.8	55.3	54.1	54.7
0.90	54.3	54.2	55.3	54.4
0.95	54.6	54.2	**55.8**	53.7
0.99	53.9	55.1	54.1	53.6

The effect of incorporating feature pyramid networks (FPNs) [59] to enhance multiscale feature representations is also examined. With the same ResNet-50 backbone, adding FPN improves performance by 0.7%, indicating that multiscale features facilitate the construction of more robust prototypes and enhance feature-level domain alignment. Notably, compared to merely increasing network depth, introducing multiscale enhancement strategies, like FPN, plays a more critical role in improving prototype quality.

Finally, leveraging a more powerful ConvNeXt-FPN as the larger teacher leads to the best mAP of 55.8%, further validating that teacher models with stronger representation capability can provide a more robust semantic foundation for prototype contrastive learning, thereby enhancing cross-domain alignment performance.

4) Effectiveness of Initial Base Confidence Threshold θ_{Base} and Base Momentum m_{Base}: In this section, a systematic analysis of two key hyperparameters in the CAPU module, the initial confidence threshold θ_{base} and the initial momentum m_{base}, is performed to determine the optimal parameter combination.

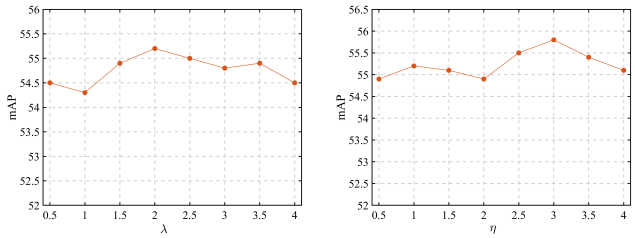
A low θ_{base} introduces many low-quality samples in the early stage, contaminating prototype construction, while a high value limits early samples but ensures that the initial prototypes are formed from high-quality features. Similarly, a low m_{base} enables fast adaptation but is prone to noise, whereas a high value maintains smooth updates and stable class centers. However, overly large values make learning conservative and hinder adaptation. Since both parameters are dynamically adjusted during training, selecting higher initial values helps form stable, high-quality prototypes at the start, which facilitates subsequent adaptation. As shown in Table VII, the combination of $\theta_{base} = 0.95$ and $m_{base} = 0.99$ achieves the best performance. This setting ensures reliable prototypes through high-quality samples and stable updates in the early stage, and with the aid of the dynamic adjustment mechanism, progressively optimizes feature representations to enhance cross-domain alignment. Further increasing either parameter leads to performance degradation, indicating that overconservative filtering and updates suppress effective domain adaptation.

5) Effectiveness of the Confidence Threshold θ_{Conf}: An ablation analysis is conducted on the confidence threshold θ_{conf}, which controls the switching between soft and hard labels when predictions are consistent, to investigate the sensitivity of the HPL mechanism to different threshold choices. As shown in Table VIII, an excessively high threshold filters out a large number of samples and limits effective supervision, while a low threshold introduces noisy hard labels, both resulting in degraded performance. Nevertheless, the performance remains relatively stable when θ_{conf} varies from 0.7 to 0.9, indicating that the HPL mechanism is not overly sensitive to this parameter within a reasonable range. The best performance is achieved at $\theta_{conf} = 0.8$, providing a favorable balance between label reliability and sample coverage.

TABLE VIII

PERFORMANCE OF HPL WITH DIFFERENT CONFIDENCE THRESHOLDS θ_{conf}

θ_{conf}	0.65	0.7	0.75	0.8	0.85	0.9	0.95
mAP	54.3	55.1	55.3	**55.8**	55.5	54.8	53.9



(a) (b)

Fig. 4. Ablation study with various hyperparameters. Specific performance of the model at different (a) λ values and (b) η values.

6) Effectiveness of Hyperparameters λ and η: To investigate the influence of more refined hyperparameter settings on model performance, the unsupervised weight λ of loss $\mathcal{L}_{det}$ and the contrastive weight η of loss $\mathcal{L}_{total}$ are analyzed, with the experimental results shown in Fig. 4.

Regarding λ, which balances the supervised and unsupervised detection losses, the results in Fig. 4(a) show that the best performance is achieved when $\lambda = 2.0$. A too-small value reduces the contribution of the unsupervised branch, whereas an excessively large value amplifies the effect of noisy pseudolabels. Nevertheless, performance remains within a narrow fluctuation range of less than 0.4% mAP when λ lies between 1.5 and 3.0, demonstrating good robustness.

As for η, which balances the detection loss and the prototype contrastive loss, the results in Fig. 4(b) show that the best performance is achieved when $\eta = 3.0$. Insufficient values weaken the discriminative power gained from contrastive learning, while overly large values can make it dominate the training process, hindering detection optimization. Choosing an appropriate η allows the model to maintain detection precision while effectively improving feature alignment across domains.

F. Visualization and Analysis

1) Detection Results: The qualitative detection results of DTPC across all domain adaptation scenarios are presented, along with comparisons to the ground truth, source-only, 2PCNet, AT, and HT. As shown in Fig. 5(a), for the NWPU

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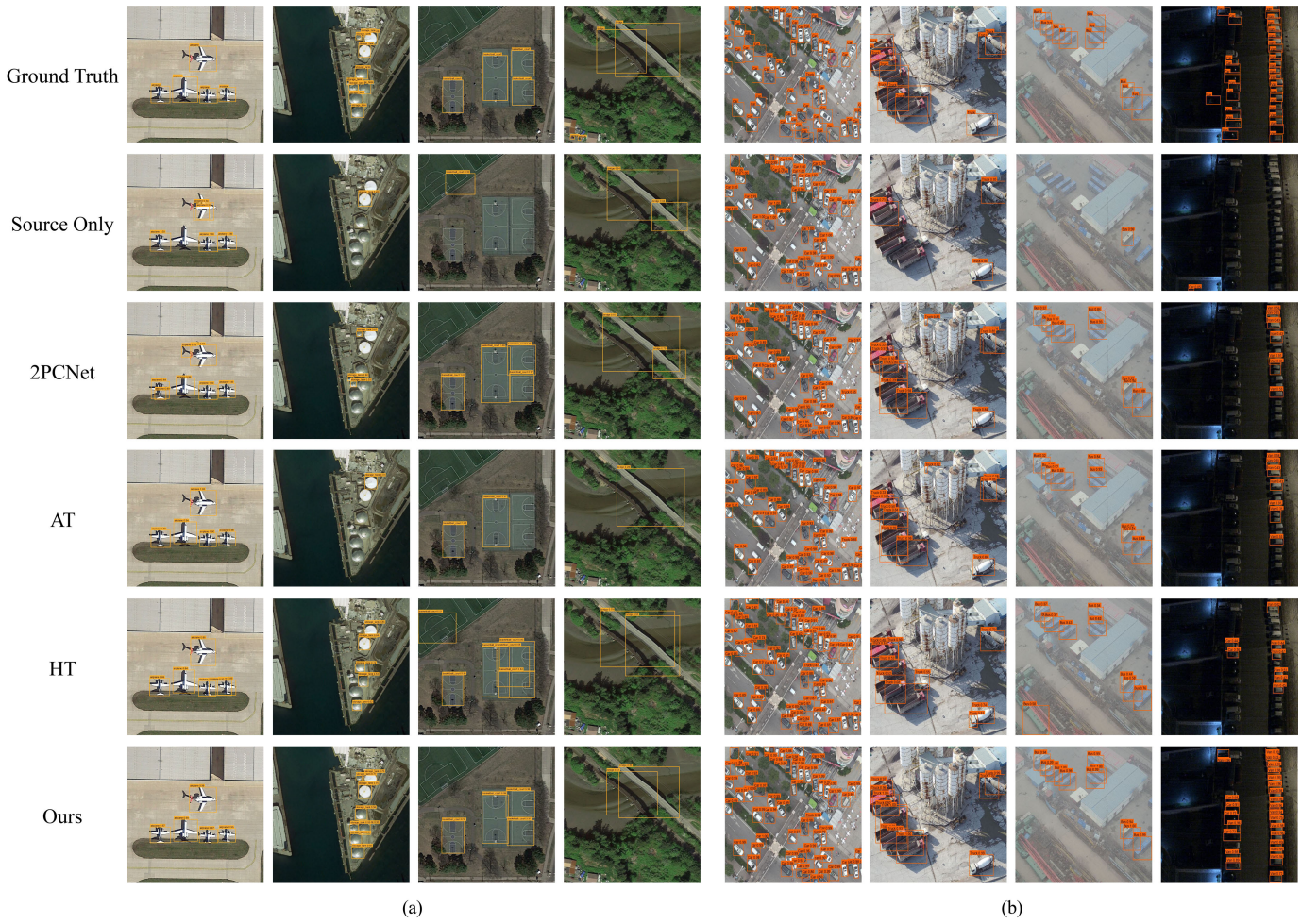


Fig. 5. Qualitative detection results on two domain adaptation scenarios. (a) NWPU VHR-10 → DIOR*. (b) VisDrone* → DroneVehicle*. (a) and (b) (Rows from top to bottom) Visualized detection results of ground truth, source-only, 2PCNet, AT, HT, and our proposed DTPC.

VHR → DIOR* adaptation task, DTPC consistently produces more accurate detections across all categories. In the visual results shown in the first, second, and fourth rows, DTPC precisely delineates object-bounding boxes. Especially in the second row, our method shows higher recall rates for less obvious features and minority categories. As shown in Fig. 5(b), in domain adaptation scenarios with more complex environmental changes, DTPC maintains excellent detection accuracy and recall rates in both foggy and low-light conditions, accurately describing bounding boxes and identifying challenging objects that may be overlooked by other methods. The results in the fourth column also confirm its enhanced performance on minority classes. In contrast, other domain adaptation methods achieve only limited improvements. These observations highlight the strong generalization capability of DTPC. The qualitative results are consistent with the quantitative evaluations, demonstrating that DTPC achieves superior performance and generalization across multiple widely adopted domain adaptation settings in remote sensing images.

2) Feature Visualization: The object features of ten classes extracted from the DIOR* dataset are visualized using the t-distributed stochastic neighbor embedding (t-SNE) method, as shown in Fig. 6. The results indicate that the baseline detector tends to confuse certain feature points, leading to closely clustered interclass distributions. In contrast, our proposed DTPC significantly enhances interclass separability and intraclass compactness in the feature space, thereby improving the cross-domain performance of the detector.

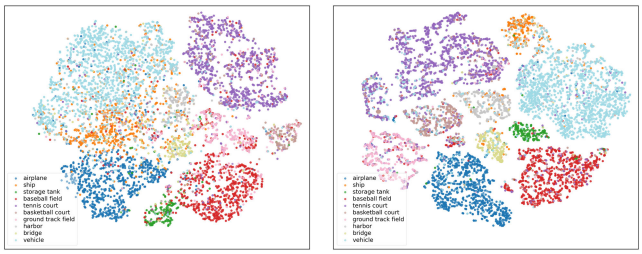


Fig. 6. t-SNE visualization of object features for ten object classes from the DIOR* dataset. Our method enhances interclass separability and intraclass compactness compared to the baseline method. (a) Baseline detector. (b) DTPC.

V. Conclusion

In this article, a DTPC framework is proposed for UDA-OD in remote sensing images. DTPC employs a frozen larger

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teacher and an EMA-updated smaller teacher to collaboratively guide the student network. The larger teacher extracts semantically rich features to construct class prototypes as domain-invariant anchors for contrastive learning, while the smaller teacher offers fine-grained and robust supervision. To enhance prototype quality under class imbalance, the CAPU mechanism dynamically adjusts the update strategy based on class frequency. Furthermore, the HPL mechanism adaptively fuses soft and hard labels through class-consistency constraints, improving both the quality and reliability of pseudolabels. Extensive experiments conducted in various remote sensing domain-adaptive scenarios show that DTPC achieves superior performance, validating the effectiveness and generalization ability of the method. In future work, object detection under source-free and multisource domain adaptation settings will be further investigated to enhance the adaptability and robustness of detection models in more complex and dynamic environments.

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